

WEEK 3 – PROMPT ENGINEERING PORTFOLIO

Student Performance Analysis Using AI Prompt Engineering

1. Objective

The objective of this activity was to understand how prompt quality and specificity affect AI-generated responses. Three prompts with increasing levels of detail were used to generate Python code for analyzing a student performance dataset.

2. Prompt 1 – Basic Prompt

Prompt

Write Python code to analyze a student performance dataset.

AI Generated Response

```
import pandas as pd

import matplotlib.pyplot as plt


df = pd.read_csv("student_performance.csv")


print(df.head())

print(df.info())

print(df.describe())


df.hist(figsize=(10, 6))

plt.tight_layout()

plt.show()
```

Evaluation

This prompt is simple and relevant, but it provides very little information about the required analysis. The generated response performs basic dataset inspection and visualization but does not provide specific analysis requirements.

Overall Score: 6.25/10

3. Prompt 2 – Improved Prompt

Prompt

Write Python code using Pandas and Matplotlib to analyze a student performance dataset. Calculate basic statistics, find correlations between study hours, attendance, previous score and final score, and create suitable visualizations.

AI Generated Response

```
import pandas as pd

import matplotlib.pyplot as plt


df = pd.read_csv("student_performance.csv")


print(df.head())

print(df.describe())


print(df[["Study_Hours", "Attendance",
          "Previous_Score", "Final_Score"]].corr())


plt.scatter(df["Study_Hours"], df["Final_Score"])

plt.xlabel("Study Hours")

plt.ylabel("Final Score")

plt.title("Study Hours vs Final Score")

plt.show()


plt.scatter(df["Attendance"], df["Final_Score"])

plt.xlabel("Attendance")

plt.ylabel("Final Score")

plt.title("Attendance vs Final Score")

plt.show()
```

Evaluation

This prompt is more specific because it identifies the required libraries, analysis methods, dataset attributes, and visualizations. The generated response is more focused and useful than the first response.

4. Prompt 3 – Detailed Prompt

Prompt

Act as a Python data analyst. Write beginner-friendly Python code using Pandas and Matplotlib to analyze a student performance dataset with Study_Hours, Attendance, Previous_Score, and Final_Score. Include data inspection, correlation analysis, and two visualizations. Keep the code short and suitable for Google Colab.

AI Generated Response

```
import pandas as pd

import matplotlib.pyplot as plt


# Load dataset

df = pd.read_csv("student_performance.csv")


# Data inspection

print(df.head())

print(df.describe())


# Correlation analysis

print(df[["Study_Hours", "Attendance",
          "Previous_Score", "Final_Score"]].corr())


# Study Hours vs Final Score

plt.scatter(df["Study_Hours"], df["Final_Score"])

plt.xlabel("Study Hours")

plt.ylabel("Final Score")

plt.title("Study Hours vs Final Score")

plt.show()


# Attendance vs Final Score

plt.scatter(df["Attendance"], df["Final_Score"])
```

```
plt.xlabel("Attendance")
plt.ylabel("Final Score")
plt.title("Attendance vs Final Score")
plt.show()
```

Evaluation

This prompt provides the most detailed instructions. It specifies the role, programming libraries, dataset columns, analysis requirements, visualization requirements, and execution environment. Therefore, it produced the most complete and focused response.

Overall Score: 9.5/10

5. Prompt Evaluation Sheet

Prompt	Clarity	Accuracy	Relevance	Completeness	Overall
Prompt 1	6/10	7/10	7/10	5/10	6.25/10
Prompt 2	8/10	8/10	9/10	8/10	8.25/10
Prompt 3	9/10	9/10	10/10	10/10	9.5/10

6. Comparison of Prompts

Prompt 1

The prompt was short and general. It generated basic code but lacked specific instructions.

Prompt 2

The prompt included the required libraries, analysis methods, dataset attributes, and visualizations. Therefore, the response was more targeted.

Prompt 3

The prompt provided detailed context, constraints, required tools, dataset columns, and expected outputs. It produced the most complete and useful response.

7. Key Learnings

- Clear prompts produce more relevant AI responses.
- Providing context helps the AI understand the task better.
- Specifying tools and libraries improves the generated code.
- Defining expected outputs makes the response more useful.

- Detailed prompts reduce the need for repeated corrections.
 - AI-generated code should always be reviewed and tested before use.
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8. Conclusion

As the prompts became more specific and structured, the AI-generated responses became more complete, relevant, and useful. This activity demonstrated the importance of clear instructions, context, constraints, and expected output when using AI tools. Effective prompt engineering can improve the quality and reliability of AI-assisted coding tasks.